

Poincaré-Covariant Quantum Operations as a Double-Coset Hypergroup

Candidate solution packet for arXiv:2303.15171, Problem 5.6

27 August 2026

Abstract

Giorgetti asks whether the relative quantum operations of a Haag-dual full Poincaré-covariant subnet $\mathcal{B} \subset \mathcal{A}$ form the double-coset hypergroup $G//H$ supplied by Doplicher–Roberts reconstruction and subsystem classification. We prove the assertion in precisely that standard regime. The local discrete-subfactor theorem identifies the maps on each double cone with the probability measures on $G//H$. Compatibility makes the local representing measures equal, and gauge–Poincaré commutation gives the converse construction. Thus

$$\mathrm{UCP}(\mathcal{A} \mid \mathcal{B}) \cong \mathrm{P}(G//H), \quad \mathrm{QuOp}(\mathcal{A} \mid \mathcal{B}) \cong G//H$$

affinely, topologically, and with the compact-hypergroup operations.

1 The source problem and its intended scope

The source is Luca Giorgetti, *Quantum Operations in Algebraic QFT*, arXiv:2303.15171 [1]. Problem 5.6 on printed page 9 reads as follows.

Lemma 5.5. *Let $(\mathcal{A}, U, \Omega)$ be a Poincaré covariant net of local observables. Then $\mathrm{Aut}(\mathcal{A}) \subset \mathrm{QuOp}(\mathcal{A})$.*

One can presumably prove the analogous results as for the chiral conformal net case. For instance: prove that $\mathrm{UCP}(\mathcal{A})$ is compact in a natural topology induced by the pointwise ultraweak operator convergence on the UCP maps ϕ_O . More importantly, in the 3+1-dimensional situation, by the classification of subsystems due to [23], [15], [17], in particular [17, Theorem 5.2], we propose the following:

Problem 5.6. Let $(\mathcal{A}, U, \Omega)$ be a Poincaré covariant net of local observables, and $\mathcal{B} \subset \mathcal{A}$ a (Haag dual, full) Poincaré covariant subnet. Then $\mathrm{QuOp}(\mathcal{A} \mid \mathcal{B}) \cong G//H$ (a *double coset hypergroup*³) for a compact group G (obtained as the Doplicher–Roberts global gauge group of the subtheory $\mathcal{B}$) and a closed subgroup $H \subset G$.

Remark 5.7. In particular, in the 3+1-dimensional context, we conjecture that $\mathrm{QuOp}(\mathcal{A})$ is always compact and a hypergroup (as close as possible to a group, in some sense).

³If H is a normal subgroup of G , then $G//H$ is a group (the quotient group), otherwise it is a prototypical example of compact hypergroup (a double quotient hypergroup), see, e.g., [10, Chapter 1.5].

In transcription: if $(\mathcal{A}, U, \Omega)$ is a Poincaré-covariant net of local observables and $\mathcal{B} \subset \mathcal{A}$ is a Haag-dual full Poincaré-covariant subnet, prove

$$\mathrm{QuOp}(\mathcal{A} \mid \mathcal{B}) \cong G//H,$$

where G is the Doplicher–Roberts compact gauge group of $\mathcal{B}$ and $H < G$ is closed.

The notation in the problem already invokes Doplicher–Roberts reconstruction and the Carpi–Conti subsystem classification. Those theorems have standing hypotheses beyond the short list in the survey’s Definition 5.1. We therefore state the exact structural conclusion that the proof uses. This both covers the intended classified setting and prevents those extra hypotheses from being silently assumed. In particular, the result applies whenever the classification cited immediately before Problem 5.6 supplies

$$\mathcal{B} = \mathcal{F}^G \subset \mathcal{F}^H = \mathcal{A}, \quad (1)$$

with its standard local irreducibility, discreteness, and locality properties for the type III inclusions. Carpi–Conti obtain precisely this fixed-point description under their standing AQFT assumptions [3, Theorem 5.2]; compare the discussion in [2, Example 9.26].

2 Idea of the proof

For a fixed double cone O , the hard operator-algebraic work is already contained in the local theorem of Bischoff–Del Vecchio–Giorgetti: all UCP maps on $\mathcal{F}(O)^H$ fixing $\mathcal{F}(O)^G$ are uniquely parametrized by probability measures on $G//H$ [2, Section 9.2, Proposition 9.7].

The apparent global obstruction is that a coherent family could conceivably choose a different measure μ_O for every O . If $O_1 \subset O_2$, however, compatibility identifies the restrictions of the two maps on $\mathcal{A}(O_1)$. The local parametrization at O_1 is injective, so $\mu_{O_1} = \mu_{O_2}$. Any two double cones are contained in a third; hence one measure parametrizes the entire family. Conversely, averaging the global gauge automorphisms against one fixed bi- H -invariant measure is automatically compatible and vacuum preserving. Internal gauge symmetries commute with U , so this family is Poincaré covariant. Taking extreme points changes the probability simplex into its Dirac masses, namely $G//H$ itself.

3 The local parametrization

Let $K := G//H = H \backslash G / H$, let $q : G \rightarrow K$ be the quotient map, and let m_H be normalized Haar measure on H . A probability measure μ on K has the canonical bi- H -invariant lift

$$\tilde{\mu} := \int_K m_H * \delta_g * m_H d\mu(HgH) \in \mathcal{P}(G). \quad (2)$$

The right-hand side is independent of the representative g .

Write $\alpha : G \rightarrow \text{Aut}(\mathcal{F})$ for the gauge action. For a double cone O and $x \in \mathcal{A}(O) = \mathcal{F}(O)^H$, set

$$\Theta_O(\mu)(x) := \int_G \alpha_g(x) d\tilde{\mu}(g). \quad (3)$$

Bi- H -invariance makes the output H -fixed, while every element of $\mathcal{B}(O) = \mathcal{F}(O)^G$ is fixed pointwise. Proposition 9.7 of [2], applied to

$$\mathcal{F}(O)^G \subset \mathcal{F}(O)^H,$$

states that

$$\Theta_O : \mathcal{P}(K) \longrightarrow \{\psi : \mathcal{A}(O) \rightarrow \mathcal{A}(O) : \psi \text{ is normal UCP and } \psi|_{\mathcal{B}(O)} = \text{id}\} \quad (4)$$

is an affine homeomorphism. In this local discrete setting the maps on the right are automatically faithful, vacuum preserving, and vacuum-adjointable; see the UCP characterization preceding the compact-hypergroup construction in [2, Sections 6–7]. The same proposition identifies composition and vacuum adjoint with the double-coset convolution and involution.

Because the gauge action is a net action, for $O_1 \subset O_2$ one has the crucial naturality identity

$$\Theta_{O_2}(\mu)|_{\mathcal{A}(O_1)} = \Theta_{O_1}(\mu). \quad (5)$$

4 The global theorem

Equip $\text{UCP}(\mathcal{A} | \mathcal{B})$ with the pointwise ultraweak topology used in the source: its members are compatible, vacuum-preserving, Poincaré-covariant families of normal faithful UCP maps that fix $\mathcal{B}$ pointwise.

Theorem 1. *Assume the fixed-point realization (1), where G acts faithfully and minimally by unbroken internal symmetries of the complete field net $\mathcal{F}$, $H < G$ is closed, and each inclusion $\mathcal{F}(O)^G \subset \mathcal{F}(O)^H$ has the irreducible local discrete type III properties used in Proposition 9.7 of [2]. Then*

$$\Theta : \text{P}(G//H) \longrightarrow \text{UCP}(\mathcal{A} | \mathcal{B}), \quad \Theta(\mu) := \{\Theta_O(\mu)\}_O, \quad (6)$$

is an affine homeomorphism. It intertwines convolution and involution. Consequently

$$\text{QuOp}(\mathcal{A} | \mathcal{B}) \cong G//H$$

as a compact hypergroup.

Proof. We prove existence, surjectivity, topology, and extremality in turn.

The integral construction gives a global family. Equation (3) is normal, unital, and completely positive. Gauge automorphisms preserve the vacuum state, hence so does their average. The local vacuum state is faithful, so a vacuum-preserving positive map cannot annihilate a nonzero positive element; thus the map is faithful. It fixes $\mathcal{B}(O)$ pointwise. Naturality (5) gives compatibility.

The action of the unbroken internal gauge group commutes with the Poincaré representation:

$$\text{Ad } U(p) \circ \alpha_g = \alpha_g \circ \text{Ad } U(p) \quad (p \in \mathcal{P}, g \in G).$$

Using this identity under the integral gives

$$\text{Ad } U(p) \circ \Theta_O(\mu) \circ \text{Ad } U(p)^{-1} = \Theta_{pO}(\mu),$$

so $\Theta(\mu) \in \text{UCP}(\mathcal{A} | \mathcal{B})$.

Every global family has one representing measure. Let $\Phi = \{\phi_O\}_O \in \text{UCP}(\mathcal{A} | \mathcal{B})$. By the local affine homeomorphism (4), for each O there is a unique $\mu_O \in \text{P}(K)$ such that

$$\phi_O = \Theta_O(\mu_O).$$

If $O_1 \subset O_2$, compatibility and (5) give

$$\Theta_{O_1}(\mu_{O_1}) = \phi_{O_1} = \phi_{O_2}|_{\mathcal{A}(O_1)} = \Theta_{O_1}(\mu_{O_2}).$$

Injectivity of Θ_{O_1} yields $\mu_{O_1} = \mu_{O_2}$. Given arbitrary double cones O_1, O_2 , choose a double cone O_3 containing both. The previous argument gives $\mu_{O_1} = \mu_{O_3} = \mu_{O_2}$. Thus all μ_O equal one measure μ , and $\Phi = \Theta(\mu)$. This proves surjectivity; injectivity follows already from any single Θ_O .

Topology and operations. For every O , the coordinate map Θ_O is continuous, so Θ is continuous for the product point-ultraweak topology. Fixing one double cone O_0 , the inverse is the

composition of evaluation at O_0 with $\Theta_{O_0}^{-1}$, hence is continuous. Therefore (6) is an affine homeomorphism. The local result gives

$$\Theta_O(\mu * \nu) = \Theta_O(\mu) \circ \Theta_O(\nu), \quad \Theta_O(\mu^*) = \Theta_O(\mu)^\sharp.$$

Equality in every double cone proves the corresponding global identities.

Extreme points. The extreme points of $P(K)$ are exactly the Dirac measures, so the extreme points of the relative convex set are $\Theta(\delta_k)$, $k \in K$. It remains only to match this with the source convention

$$\text{QuOp}(\mathcal{A} \mid \mathcal{B}) = \text{QuOp}(\mathcal{A}) \cap \{\Phi : \Phi|_{\mathcal{B}} = \text{id}\}.$$

The relative set is a face of $\text{UCP}(\mathcal{A})$. Indeed, suppose $\Phi = t\Phi_1 + (1-t)\Phi_2$, $0 < t < 1$, and Φ fixes $\mathcal{B}$. For every unitary $u \in \mathcal{B}(O)$,

$$u = t\phi_{1,O}(u) + (1-t)\phi_{2,O}(u).$$

The two terms on the right are contractions, while a unitary is an extreme point of the unit ball of a unital C^* -algebra. Hence $\phi_{1,O}(u) = \phi_{2,O}(u) = u$. Since every element of a unital C^* -algebra is a linear combination of unitaries, both Φ_i fix $\mathcal{B}$. Thus extremes of the relative set are precisely its elements that are extreme in $\text{UCP}(\mathcal{A})$, completing the identification with K . $\square$

Corollary 1 (Problem 5.6 in the classified AQFT regime). *Under the standing Doplicher–Roberts and Carpi–Conti hypotheses invoked in Problem 5.6, its proposed conclusion holds:*

$$\text{QuOp}(\mathcal{A} \mid \mathcal{B}) \cong G//H.$$

In particular, it is compact and carries the asserted double-coset hypergroup structure.

Proof. The cited subsystem classification supplies (1), with G the Doplicher–Roberts gauge group of $\mathcal{B}$ and $H < G$ closed [3, Theorem 5.2]. The associated local fixed-point subfactors are the double-coset examples of [2, Section 9.2 and Example 9.26]. Apply Theorem 1. $\square$

5 Verification, limitations, and novelty check

The proof has no numerical component. Its only substantive external inputs are (a) the fixed-point realization from subsystem classification and (b) the local affine homeomorphism of Proposition 9.7 in [2]. Once these are in place, the new local-to-global step uses only compatibility, injectivity, and the fact that double cones are directed. The face argument above also checks explicitly that taking relative extreme points agrees with the source definition.

The scope is important. If one reads only the minimal axioms in the survey’s definition of a local net, without the standard assumptions required for Doplicher–Roberts reconstruction and the Carpi–Conti classification, the fixed-point presentation need not be available and the group G asserted in the problem may not be defined. The packet makes no claim for that broader reading; it proves the problem in the very classified regime named in its statement and preceding paragraph.

A bounded novelty search on 27 August 2026 covered the run’s local indexes and live exact-title, exact-phrase, arXiv-id, author/keyword, and “ $\text{QuOp}(\mathcal{A} \mid \mathcal{B})$ ” searches. It found the source paper, the conformal-net antecedent, and the local-subfactor theorem, but no later paper claiming the Poincaré-covariant global result. This is evidence for novelty, not an exhaustive literature certification.

The recommended human check is the precise transfer from the standing hypotheses intended by Problem 5.6 to the local hypotheses of Proposition 9.7. The accompanying `verification.md` gives a line-by-line dependency audit.

References

- [1] L. Giorgetti, *Quantum Operations in Algebraic QFT*, arXiv:2303.15171 (2023), Problem 5.6.
- [2] M. Bischoff, S. Del Vecchio, and L. Giorgetti, *Compact hypergroups from discrete subfactors*, J. Funct. Anal. **281** (2021), 109004; arXiv:2007.12384.
- [3] S. Carpi and R. Conti, *Classification of subsystems for graded-local nets with trivial superselection structure*, Comm. Math. Phys. **253** (2005), 423–449; arXiv:math/0312033.